

Bayesian Learning

Lecture 6 - Large sample approximations. Classification.

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Lecture overview

- **Classification**
- **Normal approximation** of posterior
- **Logistic regression** - demo in R

Classification with logistic regression

- Regression with **binary response** ($y = 0$ or 1).
- **Logistic regression**

$$\Pr(y_i = 1 | \mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})}$$

- **Likelihood** for independent observations

$$p(\mathbf{y} | \mathbf{X}, \boldsymbol{\beta}) = \prod_{i=1}^n \frac{(\exp(\mathbf{x}_i^\top \boldsymbol{\beta}))^{y_i}}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})}$$

- **Probit regression**

$$\Pr(y_i = 1 | \mathbf{x}_i) = \Phi(\mathbf{x}_i^\top \boldsymbol{\beta})$$

Poisson regression and beyond

■ Poisson regression

$$y_i | \mathbf{x}_i \stackrel{\text{ind}}{\sim} \text{Pois}(\exp(\mathbf{x}_i^\top \boldsymbol{\beta}))$$

■ Gamma regression, exponential regression, ...

■ Generalized linear models

■ The posterior distribution is generally intractable.

■ Three approaches:

- ▶ **Numerical integration.** Works only for a few parameters.
- ▶ **Posterior simulation.** Gibbs sampling, MCMC. Lecture 7-9.
- ▶ **Posterior approximation.** Now!

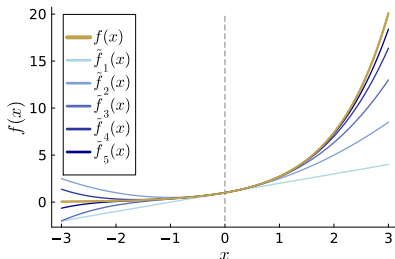
Taylor approximation

- **Taylor approximation** of the function $f(x)$ around $x = a$

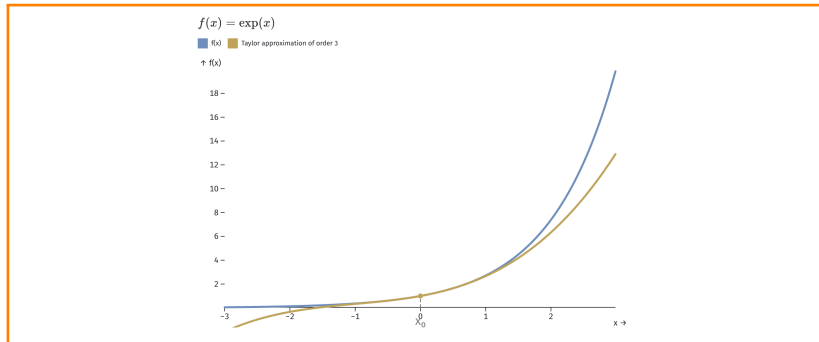
$$f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f^{(3)}(a)(x-a)^3 + \dots$$

- Taylor approximation of $f(x) = \exp(x)$ around $x = 0$

$$\exp(x) \approx \sum_{k=0}^K \frac{x^k}{k!}$$



Interactive - Taylor approximation



Normal approximation of likelihood

- **Taylor expansion of log-likelihood** around the MLE $\theta = \hat{\theta}$:

$$\begin{aligned}\ln p(\mathbf{x}|\theta) &= \ln p(\mathbf{x}|\hat{\theta}) + \frac{\partial \ln p(\mathbf{x}|\theta)}{\partial \theta} \Big|_{\theta=\hat{\theta}} (\theta - \hat{\theta}) \\ &\quad + \frac{1}{2!} \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta^2} \Big|_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 + \dots\end{aligned}$$

- From the definition of the MLE:

$$\frac{\partial \ln p(\mathbf{x}|\theta)}{\partial \theta} \Big|_{\theta=\hat{\theta}} = 0$$

- In **large samples**

$$\begin{aligned}\ln p(\mathbf{x}|\theta) &\approx \ln p(\mathbf{x}|\hat{\theta}) + \frac{1}{2} \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta^2} \Big|_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 \\ &= \ln p(\mathbf{x}|\hat{\theta}) - \frac{1}{2} J_{\mathbf{x}}(\hat{\theta}) (\theta - \hat{\theta})^2\end{aligned}$$

using the definition of the **observed information**

$$J_{\mathbf{x}}(\hat{\theta}) = - \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta^2} \Big|_{\theta=\hat{\theta}}$$

Normal approximation of likelihood

- So, taking exp on both sides:

$$p(\mathbf{x}|\theta) \approx p(\mathbf{x}|\hat{\theta}) \exp \left(-\frac{1}{2} \mathbf{J}_{\mathbf{x}}(\hat{\theta})(\theta - \hat{\theta})^2 \right)$$

- Likelihood looks like a $N(\hat{\theta}, \mathbf{J}_{\mathbf{x}}^{-1}(\hat{\theta}))$ density in large samples.
- Multi-parameter θ : Likelihood $\approx N(\hat{\theta}, \mathbf{J}_{\mathbf{x}}^{-1}(\hat{\theta}))$
- Multiparameter **observed information matrix**

$$\mathbf{J}_{\mathbf{x}}(\hat{\theta}) = -\frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta \partial \theta^T} \Big|_{\theta=\hat{\theta}}$$

- Example: $\theta = (\theta_1, \theta_2)^T$

$$\frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta \partial \theta^T} = \begin{pmatrix} \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta_1^2} & \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta_1 \partial \theta_2} \\ \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta_1 \partial \theta_2} & \frac{\partial^2 \ln p(\mathbf{x}|\theta)}{\partial \theta_2^2} \end{pmatrix}$$

Normal posterior approximation

- We can do the same Taylor approximation on log posterior

$$\log p(\boldsymbol{\theta}|\mathbf{x}) = \log p(\mathbf{x}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) - \log p(\mathbf{x})$$

- **Approximate normal posterior** in large samples

$$\boldsymbol{\theta}|\mathbf{x} \stackrel{\text{approx}}{\sim} N(\tilde{\boldsymbol{\theta}}, J_{\mathbf{x}}^{-1}(\tilde{\boldsymbol{\theta}}))$$

- $\tilde{\boldsymbol{\theta}} = \arg \max p(\boldsymbol{\theta}|\mathbf{x})$ is the posterior mode and
- $J_{\mathbf{x}}(\tilde{\boldsymbol{\theta}})$ is now the **observed posterior information**.
- Important: sufficient with proportional form

$$\log p(\boldsymbol{\theta}|\mathbf{x}) = \log p(\mathbf{x}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$$

Normal posterior approximation

Normal posterior approximation

The posterior can in large samples be approximated by

$$\boldsymbol{\theta}|\mathbf{y} \stackrel{\text{a}}{\sim} \text{N}\left(\tilde{\boldsymbol{\theta}}, J_{\mathbf{y}}^{-1}(\tilde{\boldsymbol{\theta}})\right)$$

where $\tilde{\boldsymbol{\theta}}$ is the posterior mode and

$$J_{\mathbf{y}}(\tilde{\boldsymbol{\theta}}) = -\frac{\partial^2 \ln p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} \Big|_{\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}}$$

is the $d \times d$ observed posterior information matrix at $\tilde{\boldsymbol{\theta}}$.

Example: gamma posterior

- **Poisson model:** $\theta|y_1, \dots, y_n \sim \text{Gamma}(\alpha + \sum_{i=1}^n y_i, \beta + n)$

$$\log p(\theta|y_1, \dots, y_n) \propto (\alpha + \sum_{i=1}^n y_i - 1) \log \theta - \theta(\beta + n)$$

- First derivative of log density

$$\frac{\partial \ln p(\theta|\mathbf{y})}{\partial \theta} = \frac{\alpha + \sum_{i=1}^n y_i - 1}{\theta} - (\beta + n)$$

$$\tilde{\theta} = \frac{\alpha + \sum_{i=1}^n y_i - 1}{\beta + n}$$

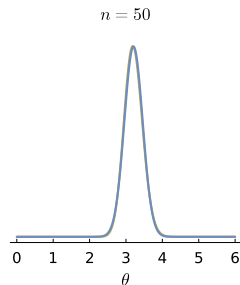
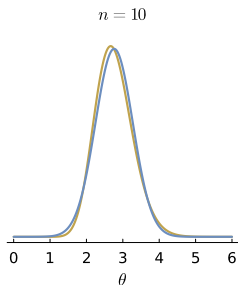
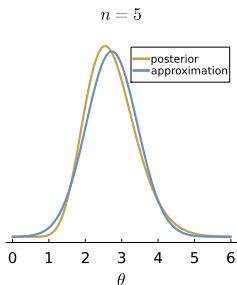
- Second derivative at mode $\tilde{\theta}$

$$\frac{\partial^2 \ln p(\theta|\mathbf{y})}{\partial \theta^2} \Big|_{\theta=\tilde{\theta}} = -\frac{\alpha + \sum_{i=1}^n y_i - 1}{\left(\frac{\alpha + \sum_{i=1}^n y_i - 1}{\beta + n}\right)^2} = -\frac{(\beta + n)^2}{\alpha + \sum_{i=1}^n y_i - 1}$$

- **Normal approximation**

$$\mathcal{N}\left[\frac{\alpha + \sum_{i=1}^n y_i - 1}{\beta + n}, \frac{\alpha + \sum_{i=1}^n y_i - 1}{(\beta + n)^2}\right]$$

Example: gamma posterior for eBay bidders data



Normal approximation of posterior

- How to compute $\tilde{\theta}$ and $J_y(\tilde{\theta})$?
- Standard **optimization routines** may be used. (`optim.r`).
- **Inputs:**
 - ▶ function: $\log p(\mathbf{y}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$.
 - ▶ Initial value $\boldsymbol{\theta}_0$.
- **Output:**
 - ▶ $\tilde{\theta}$
 - ▶ Hessian matrix $-J_y(\tilde{\theta})$.
- **Automatic differentiation** - efficient derivatives on computer.

Normal approximation of posterior

- **Re-parametrization** may improve normal approximation:
 - ▶ If $\theta \geq 0$ use $\phi = \log(\theta)$.
 - ▶ If $0 \leq \theta \leq 1$, use $\phi = \log\left(\frac{\theta}{1-\theta}\right)$.
- Even if the posterior of θ is approx normal, **interesting functions** of $g(\theta)$ may not be (e.g. predictions).
- But approximate posterior of $g(\theta)$ can be obtained by **simulating** from $N\left[\tilde{\theta}, J_y^{-1}(\tilde{\theta})\right]$.

Bayesian logistic regression

Logistic regression

$$\Pr(y_i = 1|\mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})} \text{ and } \Pr(y_i = 0|\mathbf{x}_i) = \frac{1}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})}$$

Odds

$$\text{Odds}(\mathbf{x}) \equiv \frac{\Pr(y = 1|\mathbf{x})}{\Pr(y = 0|\mathbf{x})} = \exp(\mathbf{x}^\top \boldsymbol{\beta}).$$

Odds ratio

$$OR_j = \frac{\text{Odds}(x_1, \dots, x_j + 1, \dots, x_p)}{\text{Odds}(x_1, \dots, x_j, \dots, x_p)} = \exp(\beta_j)$$

Likelihood

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\beta}) = \prod_{i=1}^n \frac{(\exp(\mathbf{x}_i^\top \boldsymbol{\beta}))^{y_i}}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})}$$

Normal approximation:

$$\boldsymbol{\beta}|\mathbf{y} \sim N\left(\tilde{\boldsymbol{\beta}}, J_{\mathbf{y}}^{-1}(\tilde{\boldsymbol{\beta}})\right).$$

Logistic regression - who survived the Titanic?

Logistic regression

$$\Pr(\text{survived}) = f(\text{intercept}, \text{age}, \text{sex}, \text{class})$$

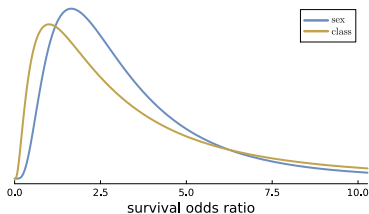
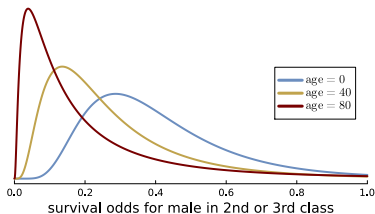
Prior

$$\beta \sim N(\mu, \Omega)$$

$$\mu = \begin{pmatrix} -1 \\ -1/80 \\ 1 \\ 1 \end{pmatrix} \quad \Omega = \begin{pmatrix} 0.25 & 0 & 0 & 0 \\ 0 & 1/(80^2) & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

age

sex and class

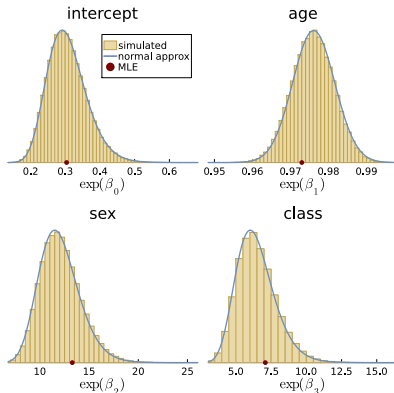


Logistic regression - who survived the Titanic?

Normal posterior approximation

$$\beta | \mathbf{y} \sim N\left(\tilde{\beta}, J_{\mathbf{y}}^{-1}(\tilde{\beta})\right).$$

- Means that the posterior of each β_j is univariate normal.
- Marginal posterior for each $\exp(\beta_j)$ is **LogNormal**.



Logistic regression - who survived the Titanic?

- Comparison with weakly informative prior $\beta \sim N(\mathbf{0}, 10^2 \mathbf{I}_p)$.

